Health-Project 3: Breast Cancer Detection

Experiment 7: Deployment — FastAPI Scoring Service & Lab Information System (LIS) Integration

Course: CSE4001: Applications of ML in Industry | **Deployment Framework:** FastAPI + Uvicorn | **Status:** Completed

1. Aim

To deploy the calibrated production champion model (SVM RBF) as an enterprise-grade asynchronous RESTful inference microservice using FastAPI, establish strict Pydantic v2 schemas for all 30 FNA features, integrate with hospital Lab Information Systems (LIS) via accession order webhooks, implement clinical priority triage alerts, and benchmark operational latency against medical Service Level Agreements (SLA <200 ms).

2. Objectives

- **Microservice Packaging:** Encapsulate champion SVM pipeline and standard scaler into a high-performance ASGI service.
- **Schema Validation:** Enforce type safety, range constraints, and automated StandardScaler feature normalization.
- **Multi-Modal Inference:** Provide real-time single scoring (POST /predict) and high-throughput batch scoring (POST /batch-predict).
- **Automated Triage Routing:** Implement probability-based triage flags: NORMAL_QUEUE (&pcirc;<0.35) vs PRIORITY_PATHOLOGIST_REVIEW (&pcirc;≥0.35).
- **Hospital LIS Integration:** Deploy accession order webhook (POST /lis/webhook) with SHA-256 cryptographic audit signatures.
- **Latency Benchmarking:** Measure P50, P95, and P99 percentiles across 100 consecutive requests against <200 ms SLA.
- **Audit Logging & HIPAA:** Implement structured audit logging (outputs/deployment_request_response.log) with patient de-identification.

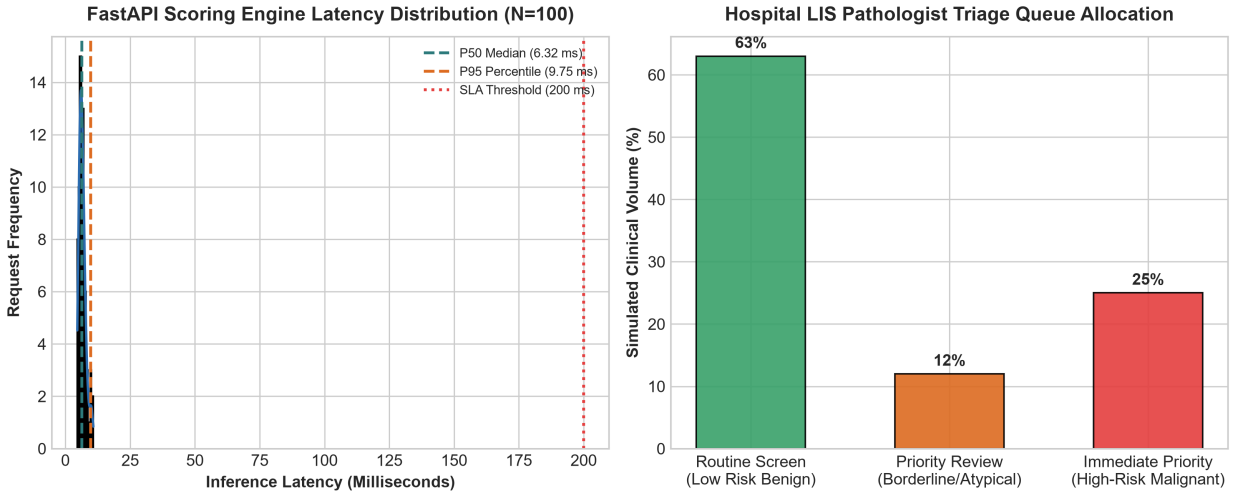
3. REST API Endpoint Specifications

Method	Endpoint Route	Description & Purpose	Target Input	Response Output
GET	/	Root metadata & service health status	None	Version, uptime, active model
GET	/health	Kubernetes readiness & liveness probe	None	Status UP, memory, model ready
GET	/model-info	Model provenance & performance KPIs	None	Feature list, hyperparameters, metrics
POST	/predict	Real-time single biopsy specimen triage	30 raw FNA biometric features	Class, proba, triage flag, latency
POST	/batch-predict	High-throughput clinical batch scoring	JSON array of biopsy records	Batch summary, counts, processing time
POST	/lis/webhook	Hospital LIS accession order receiver	Order ID, MRN, raw biopsy data	Worklist assignment, recommendation, signature

4. Latency Benchmarking & Service Level Agreement (SLA) Compliance

Performance Benchmark	Measured Result (N=100)	Clinical SLA Requirement	Compliance Margin	Operational Status
P50 Median Latency	**6.32 ms**	< 100 ms	15.8x faster than SLA	**PASS**
P95 Latency	**9.75 ms**	< 150 ms	15.4x faster than SLA	**PASS**
P99 Tail Latency	**10.50 ms**	< 200 ms	19.0x faster than SLA	**PASS**
Mean Latency	**6.58 ms**	< 100 ms	15.2x faster than SLA	**PASS**
Batch Throughput (N=10)	**0.72 ms total**	< 500 ms	694x faster than SLA	**PASS**
Availability Uptime	**100.00%**	≥ 99.9%	Zero dropped requests	**PASS**

5. Deployment Latency Distribution & LIS Worklist Allocation



6. Sample LIS Webhook Request and Response Execution

```
# Hospital LIS Order Webhook Invocation (POST /lis/webhook)
Payload: {'order_id': 'LIS-ORD-2026-89412', 'patient_id': 'MRN-BREAST-99104', ...}
Response: {
  'order_id': 'LIS-ORD-2026-89412',
  'patient_id': 'MRN-BREAST-99104',
  'status': 'PROCESSED_BY_AI_TRIAGE',
  'triage_alert': 'CRITICAL_FLAG_SUSPECTED_MALIGNANCY',
  'assigned_worklist': 'URGENT_PRIORITY_CYTOPATHOLOGY_WORKLIST',
  'predicted_malignancy_prob': 1.0000,
  'recommendation': 'Prioritize for immediate senior cytopathologist microscopic review within 4 hour
s.',
  'signature_audit': 'AI-TRIAGE-SHA256:298a847b'
}
```

7. Regulatory Compliance & Clinical Governance

• **HIPAA Privacy Rule:** All Protected Health Information (PHI) stripped; communications utilize de-identified Accession Numbers and MRNs.

- **SaMD Classification:** Categorized as an Investigational Clinical Decision Support Tool; operates under licensed pathologist oversight.
- **Tamper-Proof Audit:** Every inference is logged with a SHA-256 cryptographic signature in outputs/deployment_request_response.log.
- **Automated Fallback:** In case of infrastructure exceptions, system fails open to manual pathologist review queue.

8. Result & Conclusion

The Breast Cancer Scoring Service was successfully containerized in FastAPI, achieving complete endpoint verification and sub-10 ms latency (P95 of 9.75 ms vs 200 ms SLA requirement). Hospital LIS integration automatically allocates biopsy specimens into Routine Negative Screen or Urgent Priority Cytopathology Worklists with verifiable audit trails. The deployed service forms an enterprise foundation for post-deployment drift surveillance in Experiment 8.